MCR 3U Exam Review

Introduction to Functions

1. Determine which of the following equations represent functions. Explain. Include a graph.
   a) $3x+4y=5$ b) $x^2+y^2=4$ c) $x^2+y=4$ d) $x+y^2=0$

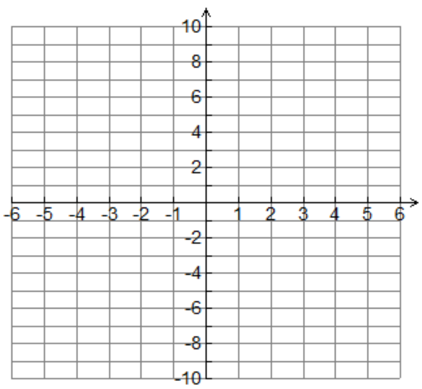
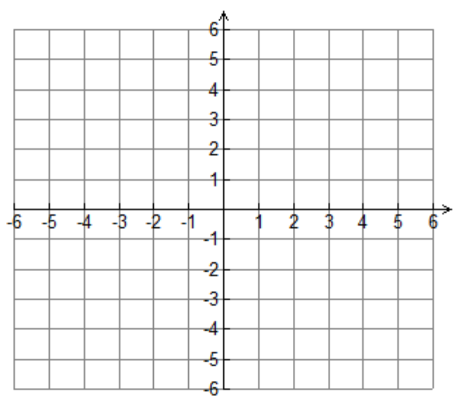
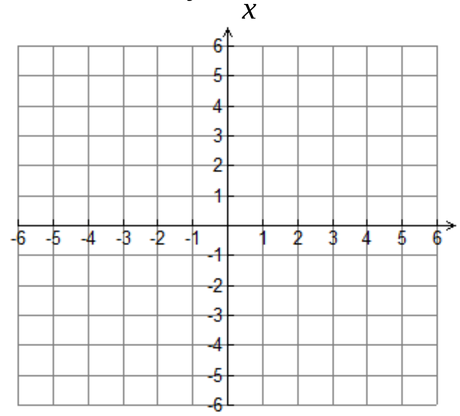
2. State the domain and range for each relation in question 1.

3. If $f(x)=3x+5$ and $g(x)=x^2+3x-5$, determine the following:
   a) $f(-3)-g(-2)$ b) $g(2x+1)-f(3x)$

4. Let $f(x)=3x^2-2x-5$. Determine the values of x for which
   a) $f(x)=0$ b) $f(x)=3$

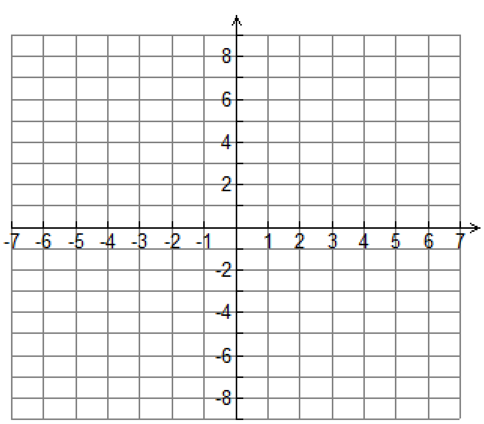
Recall the base graphs.

$y=x^2$ $y=\sqrt{x}$ $y=\frac{1}{x}$

5. Graph $y=\dfrac{-2}{x-1}+3$. State the domain and range.

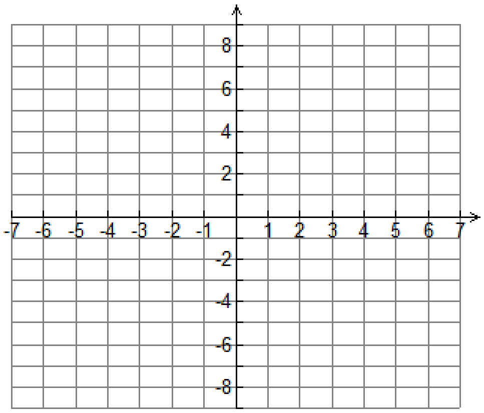
Describe how the graph can be obtained from the graph of $y=\dfrac{1}{x}$.

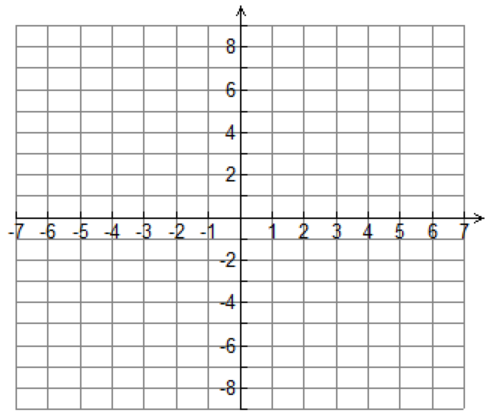


Also Try!

a) Graph $y = \dfrac{2}{x+2} - 2$

b) $y = \dfrac{-1}{x-3} + 2$



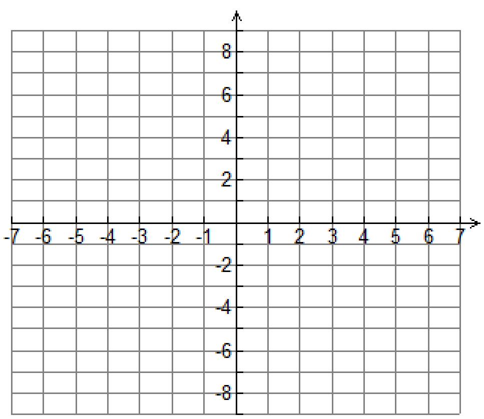


6. Given $f(x) = (x-1)^2 - 3$,

a) determine the equation of the inverse

b) graph $f(x)$ and its inverse

c) Is the inverse a function? **Explain**.
   If not, restrict the domain of $f(x)$ so that the inverse is a function.



Algebraic Expressions

1. Evaluate each of the following.

a) $\sqrt[3]{\frac{8}{27}} + 16^{-\frac{3}{4}}$ b) $\left(\sqrt[3]{5^2}\right)\left(\sqrt[3]{5}\right)$ c) $\frac{-\sqrt[3]{512}}{\sqrt[5]{-1024}}$

2. Simplify. Express each answer with positive exponents.

a) $\frac{\left(-2m^{-2}n\right)\left(5m^{-3}n^2\right)}{4m^2n^{-3}}$ b) $\frac{\left(4r^{-7}\right)\left(-2r^2\right)^5}{\left(-2r\right)^4}$ c) $\left(\frac{-25x^{-3}y^2}{15x^2y^{-1}}\right)^{-2}$

3. Simplify and state restrictions

a) $\frac{4x^2-1}{2x^2-3x-2} \times \frac{2x^2-5x+2}{4x-2}$ b) $\frac{4}{x^2-3x+2} + \frac{2}{x^2-4}$

c) $\frac{a-5}{a^2-6a+8} - \frac{a-3}{2a^2-3a-2}$ d) $\frac{8y^2+2y-1}{6y^2-y-2} \div \frac{8y-2}{6-9y}$

4. Is $(2x+3y+4z)^2 = 4x^2+9y^2+16z^2$? **Justify your response.**

5. Is $(2x-3)^3 = (3-2x)^3$? **Justify your response.**

Quadratic Functions

1. Simplify each of the following.

a) $3\sqrt{27}-2\sqrt{12}+\sqrt{108}$ b) $\sqrt{5}(3\sqrt{5}-5\sqrt{3})$ c) $(3\sqrt{3}+4\sqrt{2})(5\sqrt{3}-2\sqrt{2})$

2. Solve.

a) $\left(a+\frac{3}{4}\right)^2=\frac{9}{16}$ b) $2x^2=-18-12x$ c) $\frac{x^2}{5}=\frac{x}{30}+\frac{1}{6}$

3. Solve. Express solutions in simplest radical form.

a) $x^2+2x-1=0$ b) $3x^2-8x+2=0$

4. Find the maximum or minimum value of the function and the value of x when it occurs.

a) $y=-\frac{1}{3}x^2+2x+4$ b) $y=5x-3x^2$

5. Write a quadratic equation, in standard form, with the roots

a) $3+\sqrt{2}$ and $3-\sqrt{2}$ and that passes through the point (3, 1).

b) $-1+2\sqrt{3}$ and $-1-2\sqrt{3}$ and that passes through the point (-1, 4).

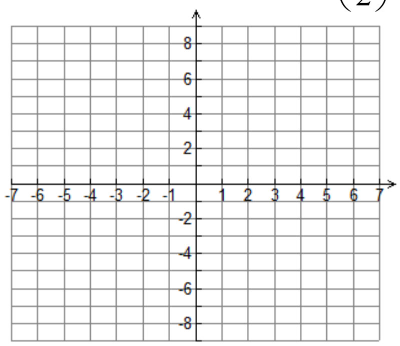
6. The sum of two numbers is 20. What is the least possible sum of their squares?

7. Two numbers have a sum of 22 and their product is 103. What are the numbers ,in simplest radical form.

MCR 3U

Exponential Functions

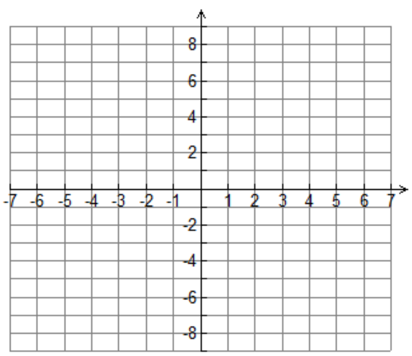
1. Graph $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ on the grid below.



a) What transformation on $y = 2^x$ will give $y = \left(\frac{1}{2}\right)^x$ as its image?

b) How are the curves alike? How are they different?

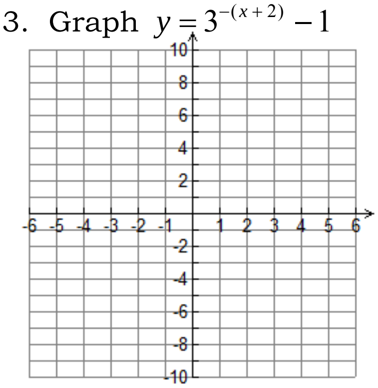
2. Graph $y = 2^x$ and $y = -2^x$ on the grid below.



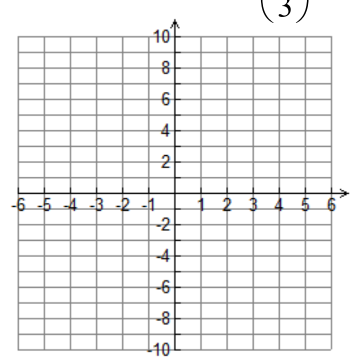
a) What transformation on $y = 2^x$ will give $y = -2^x$ as its image?

b) How are the curves alike? How are they different?

3. Graph $y = 3^{-(x+2)} - 1$



4. Graph $y = -\left(\frac{1}{3}\right)^{x-2} + 4$



5. Solve each of the following.

a) $3^{4x-1} = 27^{3x}$

b) $27^{x+1} = \left(\frac{1}{9}\right)^{2x-5}$

6. **Half-Life** A laboratory has 40 mg of iodine 131. After 24 days there are only 5 mg remaining. What is the half life of iodine 131?

Trigonometry

1. Given $\cos\theta = \frac{-4}{\sqrt{20}}$, where $0° \le \theta \le 180°$

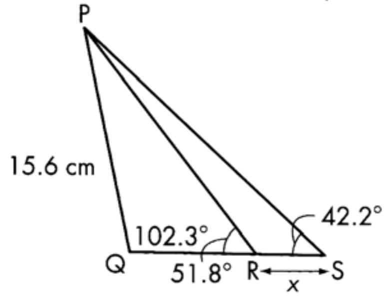
a) Find the six trigonometric ratios. Express each in simplest radial form.
b) Determine the measure of θ, to the nearest degree.

2. If $0° \le \theta \le 360°$, find the possible measure of angle θ.

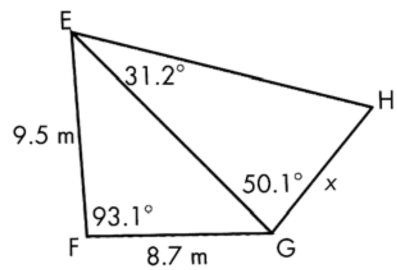
a) $\sin\theta = \frac{\sqrt{3}}{2}$ b) $\cos\theta = -\frac{1}{\sqrt{2}}$ c) $\tan\theta = -\frac{1}{\sqrt{3}}$

3. Find the length of the indicated side.

a)



b)



4. Solve each of the following triangles. Include a diagram.

a) ΔDEF, $\angle D = 61°, d = 9cm,\ f = 8cm$
b) ΔRST, $\angle S = 35°, t = 7cm, s = 5cm$
c) ΔPQR, $\angle P = 30°,\ p = 12cm, q = 24cm$

5. Prove each identity.

a) $\csc^2\theta + \sec^2\theta = (\csc^2\theta)(\sec^2\theta)$

b) $\frac{1+\cos\theta}{\sin\theta} - \frac{\sin\theta}{1-\cos\theta} = 0$

c) $\frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos^2\theta - \sin^2\theta$

6. Calculate each of the following values exactly and then simplify.

a) $\sin 135° \cos 225° - \sin^2 240° \cos(-60°)$ b) $\sin 225° \cos 315° - \tan 210° \tan 240°$

7. A carnival Ferris wheel with a radius of 5.5 m and axle 7 m above the ground, makes one complete revolution every 24 s.

a) Draw a graph to show how a person's height above the ground varies with time, starting when the person gets on to the Ferris wheel at its lowest point.
b) Find an equation for the graph.

8. Sketch one cycle of the graph of each of the following. State the domain and range of the cycle.

a) $y = 3\sin\left(\frac{\theta}{2} - 45°\right) - 2$ b) $y = -2\cos(2x + 90°) + 3$

Sequences and Series

1. A balloon filled with helium has a volume of 20 000 cm^3. The balloon loses one fifth of its helium every 24 h. What volume of helium will be in the balloon at the start of the sixth day? the seventh day?

2. An accelerating rocket rises 10 m in the first second, 40 m in the second second, and 70 m in the third second. If the arithmetic sequence continues, how high will the rocket be after 20 s?

3. The air in a hot air balloon cools as the balloon rises. If the air is not reheated, the balloon rises more slowly every minute. Suppose that a hot-air balloon rises 50 m in the first minute. In each succeeding minute, the balloon rises 70% as far as it did in the previous minute. How far does the balloon rise in 7 min, to the nearest metre?

4. In a geometric sequence, $a = 13$, $t_3 = 208$ and the sum of all terms of the series is 42 601. How many terms are in the series?
5. Determine the first and last terms of an arithmetic series with 50 terms, a common difference of 6, and a sum of 7850.

6. Determine the value of x that makes each sequence **geometric**.
   a) $4, 8, 16, 3x + 2, \ldots$ b) $2, 6, 5x - 2, \ldots$

7. Determine the value of x that makes each sequence **arithmetic**.
   a) $x - 2, x + 2, 5, 9, \ldots$ b) $x - 4, 6, x, \ldots$

8. Expand and simplify

   a) $(x - 3y)^4$ b) $(2x - 3y)^5$ c) $\left(x - \frac{3}{x}\right)^6$

Finance

1. Gail's grandfather saved for a trip by depositing $400 at the end of each month for 18 months. The account earns 4.8% per annum, compounded monthly. How much will be in the account when the last deposit is made?

2. Leila is borrowing $65 000 for 4 years. She is deciding between a loan at 6.95% per annum, compounded monthly, and a loan at 7% per annum, compounded annually.
a) Predict which loan is the better deal.
b) How much interest is paid on the loan that is the better deal?

3. Marcia wants to receive $10 000 every 6 months, for 3 years for living expenses when she goes back to school, starting 6 months from now. How much money must she invest now at 4.25% per annum, compounded semi-annually.